

Categories, Proofs, and Processes:

Lecture 17 Adjunctions

Wira Azmoon Ahmad

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7 Adjunctions

7.1 Motivating Example

Consider two non-isomorphic posets $(\mathbb{Z}, \leq)$, $(\mathbb{R}, \leq)$ with the obvious inclusion map

$$i : (\mathbb{Z}, \leq) \hookrightarrow (\mathbb{R}, \leq) := z \mapsto z$$

which preserves order

$$z \leq z' \implies i(z) \leq i(z')$$

We can't invert i , but can "do our best" Given $r \in \mathbb{R}$, find some z such that $i(z) = r$.

7.1.1 Approximate from below

$$f(r) := \text{largest } z \text{ s.t. } i(z) \leq r$$

i.e. "floor of r ". f is the right adjoint of i . (i is the left adjoint of f .)

$$i(z') \leq r \iff z' \leq f(r)$$

7.1.2 Approximate from above

$$c(r) := \text{smallest } z \text{ s.t. } r \leq i(z)$$

i.e. "ceiling of r ". c is the left adjoint of i . (i is the right adjoint of c .)

$$r \leq i(z') \iff c(r) \leq z'$$

$$\begin{array}{ccc} & f & \\ \mathbb{Z} & \begin{array}{c} \xrightarrow{\quad} \\ \dashv \\ \xrightarrow{\quad} \end{array} & \mathbb{R} \\ & c & \end{array}$$

Definition 7.1. In posets, the Galois connection is a special case of an adjunction.

7.2 Definition

Definition 7.2 (Adjunction, Def. 68, pg. 45). An adjunction from $\mathcal{C}$ to $\mathcal{D}$ is a triple (F, G, θ) , where functors F, G ,

$$\mathcal{C} \begin{array}{c} \xrightarrow{F} \\ \xleftarrow{G} \end{array} \mathcal{D}$$

and θ is a family of bijections

$$\theta_{A,B} : \mathcal{C}(A, G(B)) \cong \mathcal{D}(F(A), B)$$

F is **left adjoint** to G , G is **right adjoint** to F .

7.3 Examples

Example 7.3. Forgetful Functor on **Pre**, category of preorders and order-preserving maps.

$$U : \mathbf{Pre} \rightarrow \mathbf{Set}$$

$$U :: (P, \leq) \mapsto P, \quad f \mapsto f$$

Note 7.4. Q: Is there a canonical way to “put the order back”, such that either:

- (a) there will always be an arrow $(P, \leq') \rightarrow (P, \leq)$? “Approximating below”
- (b) there will always be an arrow $(P, \leq) \rightarrow (P, \leq')$? “Approximating above”

For (a), we can let $a \leq a$ and nothing else.

$$\implies a \leq' b \implies a = b \implies 1_P(a) \leq 1_P(b)$$

1_P (the function) is order preserving $\leq'$ is called the discrete order $\leq_d$.

For (b), we can let $a \leq' b$ for all $a, b \in P$

$$a \leq b \implies 1(a) \leq' 1(b)$$

1_P is order preserving $\leq'$ is the indiscrete order $\leq_i$

Let $D : \mathbf{Set} \rightarrow \mathbf{Pre}$

$$D(A) = (A, \leq)$$

$$D(f) = f$$

$$\forall a, b, a \leq b \implies a = b \implies f(a) \leq f(a) \implies f(a) \leq f(b)$$

Any function out of a discrete order is an order preserving map $\mathbf{Pre}((X, \leq), (Y, \leq)) = \mathbf{Set}(X, Y)$

$$\mathbf{Pre}(DA, B) = \mathbf{Set}(A, UB)$$

D is the left adjoint of U .

Let $I : \mathbf{Set} \rightarrow \mathbf{Pre}$

$$I(A) = (A, \leq)$$

$$I(f) = f$$

Any function into the indiscrete preorder is order-preserving.

$$a \leq b \implies f(a) \leq b$$

$$\mathbf{Pre}((X, \leq), (Y, \leq)) = \mathbf{Set}(X, Y)$$

$$\mathbf{Pre}(A, IB) \cong \mathbf{Set}(UA, B) \implies U \dashv I$$

I is the right adjoint of U .

$$\begin{array}{ccc} & \xleftarrow{D} & \\ \mathbf{Pre} & \xleftrightarrow[\perp]{I} & \mathbf{Set} \\ & \xleftarrow{I} & \end{array}$$

▲

Example 7.5.

$$\begin{array}{ccc} & \xrightarrow{\mathbb{R}[-]} & \\ \mathbf{Set} & \xleftrightarrow[\perp]{U} & \mathbf{Vect}_{\mathbb{R}} \\ & \xleftarrow{U} & \end{array}$$

$$\mathbb{R}[-] \dashv U$$

$$\eta_X : X \rightarrow U\mathbb{R}[X]$$

$$\eta_X :: e_i \in X \mapsto e_i \in U\mathbb{R}[X]$$

$\mathbb{R}[X]$ is the vector "spanned" by X . If $X = \{e_1, e_2, e_3\}$, then $\mathbb{R}[X] \cong \mathbb{R}^3$

$$\mathbb{R}[f] : \mathbb{R}[X] \rightarrow \mathbb{R}[Y]$$

$$\mathbb{R}[f](\sum_i r_i e_i) := \sum_i r_i f(e_i)$$

To find the linear map, just find where the basis elements go.

▲

7.4 Unit

$$\begin{array}{ccc} & \xrightarrow{F} & \\ \mathcal{C} & \xleftrightarrow[\perp]{G} & \mathcal{D} \\ & \xleftarrow{G} & \end{array}$$

is an adjunction iff $\forall X \in \text{Ob}(\mathcal{C})$, there exists a unit

$$\eta_X : X \rightarrow GF(X)$$

such that

$$\forall f : X \rightarrow G(Y), \exists ! \hat{f} : F(X) \rightarrow Y$$

where

$$\begin{array}{ccc} X & \xrightarrow{\eta_X} & GF(X) \\ & \searrow f & \downarrow G(\hat{f}) \\ & & G(Y) \end{array}$$

Example 7.6. Makes more sense as unit, unit version.

$$\begin{array}{ccc} & \xrightarrow{\mathbb{R}[-]} & \\ \mathbf{Set} & \perp & \mathbf{Vect}_{\mathbb{R}} \\ & \xleftarrow{U} & \end{array}$$

$\mathbb{R}[X]$ vector space spanned by X .

$$v \in \mathbb{R}[X], v = \sum_i r_i x_i, \quad r_i \in \mathbb{R}, x_i \in X$$

$$\mathbb{R}[f] : \mathbb{R}[X] \rightarrow \mathbb{R}[Y]$$

$$\mathbb{R}[f] \left(\sum_i r_i x_i \right) = \sum_i r_i f(x_i)$$

$$\begin{array}{ccc} b & \xrightarrow{\quad} & b \\ & \searrow & \downarrow \\ & B & \xrightarrow{\quad} U(\mathbb{R}[B]) \\ & \searrow f & \downarrow U(f) \\ & & U(V) \\ & & \downarrow \\ & & f(b) \end{array} \quad \begin{array}{ccc} \mathbb{R}[B] & & \sum_i r_i k_i \\ \hat{f} \downarrow & & \downarrow \\ V & & \sum_i r_i f(b_i) \end{array}$$

▲

Example 7.7. Makes more sense as co-unit, unit version.

$$(- \times B) : \mathbf{Set} \rightarrow \mathbf{Set}$$

$$\begin{cases} A \mapsto A \times B \\ f \mapsto f \times 1_B \end{cases}$$

$$g : X \rightarrow Y$$

$$B \Rightarrow g : B \Rightarrow X \rightarrow B \Rightarrow Y$$

$$(B \Rightarrow g)(h) := g \circ h$$

$$\begin{array}{ccc} x & \xrightarrow{\quad} & \lambda b.(x, b) \\ & \searrow & \downarrow \\ & X & \xrightarrow{\eta_X} B \Rightarrow (X \times B) \\ & \searrow f & \downarrow (f \circ -) \\ & & B \Rightarrow Y \\ & & \downarrow \\ & & f(x) = (\hat{f} \circ -)(\lambda b.(x, b)) \end{array}$$

$$\begin{aligned}
\hat{f} &: X \times B \rightarrow Y \\
\hat{f}(x, b) &= f(x)(b) \\
(\hat{f} \circ -)(\lambda b.(x, b)) &= \lambda b.\hat{f}(x, b) = \lambda b.f(x)(b) = f(x)
\end{aligned}$$

Last equality by η expansion.

▲

7.4.1 Notes

- Example: X is set, $GF(X)$ is a list.
- More useful for the left adjoints.

7.5 Bijection

$$\mathcal{D}(F(X), Y) \cong \mathcal{C}(X, G(Y))$$

7.6 Co-Unit

$$\begin{array}{ccc}
& F & \\
\mathcal{C} & \xrightarrow{\quad} & \mathcal{D} \\
& \perp & \\
& G &
\end{array}$$

is an adjunction iff $\forall X \in \text{Ob}(\mathcal{C})$, there exists a co-unit

$$\varepsilon_Y : FG(Y) \rightarrow Y$$

such that

$$\forall f : F(X) \rightarrow Y, \exists ! \tilde{f} : X \rightarrow G(Y)$$

where

$$\begin{array}{ccc}
F(X) & & \\
\downarrow F(\tilde{f}) & \searrow f & \\
FG(Y) & \xrightarrow{\varepsilon_Y} & Y
\end{array}$$

Note 7.8. Co-unit is some kind of evaluation.

Example 7.9. Makes more sense as unit, co-unit version.

$$F = \mathbb{R}[-], G = U$$

$$\begin{array}{ccc}
& \mathbb{R}[-] & \\
\mathbf{Set} & \xrightarrow{\quad} & \mathbf{Vect}_{\mathbb{R}} \\
& \perp & \\
& U &
\end{array}$$

"Forget that Y is a vector space, treat it as a set"

$$\begin{array}{ccc}
 \mathbb{R}[X] & & \\
 \downarrow F(\tilde{f}) & \searrow f & \\
 \mathbb{R}[U(Y)] & \xrightarrow{\varepsilon_Y} & Y \\
 \uparrow \text{wavy} & & \\
 \sum_i r_i \cdot [v_i] & \longmapsto & \sum_i r_i v_i
 \end{array}$$

$$\tilde{f}(x) = f(x)$$

▲

Example 7.10. Makes more sense as co-unit.

$$\begin{array}{c}
 (- \times B) : \mathbf{Set} \rightarrow \mathbf{Set} \\
 \left\{ \begin{array}{l} A \mapsto A \times B \\ f \mapsto f \times 1_B \end{array} \right. \\
 \mathbf{Set}(A \times B, C) \cong \mathbf{Set}(A, B \Rightarrow C) \\
 \begin{array}{ccc}
 & \xrightarrow{(- \times B)} & \\
 \mathbf{Set} & \perp & \mathbf{Set} \\
 & \xleftarrow{G=(B \Rightarrow -)} &
 \end{array} \\
 FG(C) = (B \Rightarrow C) \times B \\
 \begin{array}{ccc}
 (a, b) & & \\
 \downarrow & \searrow & \\
 (\lambda x. f(a, x), b) & \xrightarrow{\quad} & f(a, b)
 \end{array}
 \end{array}$$

$$\begin{array}{ccc}
 & A \times B & \\
 \tilde{f} \times 1_B \downarrow & \searrow f & \\
 (B \Rightarrow C) \times B & \xrightarrow{\varepsilon_C} & C
 \end{array}$$

$$\begin{array}{l}
 \tilde{f} : A \rightarrow (B \Rightarrow C) \\
 \tilde{f} :: a \mapsto \lambda x. f(a, x) \\
 \tilde{f} :: a \mapsto f(a, -)
 \end{array}$$

Notation mapping of the same thing.

▲